

## Sets Theory

A **set** is a **collection of objects**, called elements of the set. A set is usually denoted by **capital letters** A, B, C, ... and elements are denoted by **small letters**.

If  $x$  is an element in the set  $A$ , then we write  $x \in A$  and read  $x$  is an **element of** (or **belongs to**)  $A$ . Its **negation** is represented by  $\notin$ .

### Methods of Representation of a Set

#### (1) The Direct Method (Listing Method)

A set can be represented by **listing its elements** between **brackets** of the type  $\{..., ...\}$ . For **example**,  $S = \{a, b, c\}$

#### (2) The Indirect Method (Property Method)

A set can be represented by writing the **property of elements** of the set and enclose the whole description in **brackets** of the type  $\{..., ...\}$ .

**Example:**

$$S = \{x: 1 \leq x \leq 5; \ x \in \mathbb{N}\};$$

$$A = \{x: x \text{ is the Sudanese universities}\}$$

#### (3) Venn Diagrams

A Venn diagram uses overlapping **circles or other shapes** to illustrate the logical relationships between two or more sets of items.

### Types of Sets

#### (1) Finite Set

A set consisting a **finite** and **countable** number of elements.

**Example:**  $S = \{a, b, c\}$

#### (2) Infinite Set

A set consisting an **infinite** and **uncountable** number of elements.

**Example:**  $S = \{a, b, c, \dots\}$

#### (3) Empty Set

Any set that does not contain any element is called the **empty** or **null**. The symbol used to represent an empty set is  $\{\ }$  or  $\emptyset$

#### (4) Subset

If  $A$  and  $B$  are sets and every element of  $A$  is also an element of  $B$ , then:

- $A$  is a **subset** of  $B$ , denoted by  $A \subseteq B$ , or **equivalently**,
- $B$  is a **superset** of  $A$ , denoted by  $B \supseteq A$

If  $A$  is a subset of  $B$ , but  $A$  is **not equal** to  $B$  (i.e. there exists at least one element of  $B$  which is not an element of  $A$ ), then:

- $A$  is a **proper subset** of  $B$ , denoted by  $A \subset B$ , or **equivalently**,
- $B$  is a **proper superset** of  $A$ , denoted by  $B \supset A$

**Example:**

- 1- If  $A = \{1,3,5\}$  and  $B = \{1,5\}$   
Then  $B$  is a **proper subset** of  $A$ .  $B \subset A$
- 2- If  $A = \{1,3,5\}$  and  $B = \{1,3,4,5\}$   
Then  $B$  is a **proper superset** of  $A$ .  $B \supset A$
- 3- If  $A = \{1,3,5\}$  and  $C = \{1,3,5\}$   
Then  $C$  is a **subset** of  $A$   $C \subseteq A$ .  
Also,  $C$  is a **superset** of  $A$   $C \supseteq A$   
But  $C$  it is **not a proper superset** of  $A$ , since  $C = A$ .

**(5) Universal Set**

A universal set contains all the elements of a problem under consideration. It is generally represented by the letter  $U$ .

**(6) Power Set**

The **collection of all the subsets** of a set  $A$  is called the **power set** of  $A$ , and is represented  $\mathcal{P}(A)$ .

**Example:** Find all the subsets (**trivial** and **real**) of the following:

(1)  $A = \{x, y, z\}$

**Trivial subsets** are  $A, \emptyset$

**Real subsets** are  $\{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}$

**Power set** is  $\mathcal{P}(A) = \{A, \emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}\}$

(2)  $B = \{a, b, c, d\}$

**Trivial subsets** are  $B, \emptyset$

**Real subsets** are  $\{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{a, b, c\}, \{a, b, d\}, \{b, c, d\}, \{a, c, d\}$

**Power set** is  $\mathcal{P}(B) = \{B, \emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{a, b, c\}, \{a, b, d\}, \{b, c, d\}, \{a, c, d\}\}$

**In general**, the number of **trivial** and **real** subsets (**Power set**) is  $2^n$ , where  $n$  is the **number of elements**.

**(7) Singleton Set**

A set having exactly one element.

**Example:** Find the **power set** of the singleton set  $A = \{5\}$

$$\mathcal{P}(A) = \{\emptyset, \{5\}\}$$

**Order of Set (Cardinality)**

The **order of set**  $A$  is the number of elements of this set and denoted by  $|A|$ .

**Example:** Find the order of the following sets

$$\begin{array}{lll} A = \{1,3,a,c,6\} & |A| = 5 & ; \quad |\emptyset| = 0 \\ B = \{2,r,a,a,c,c\} & |B| = 4 & ; \quad |\{\emptyset\}| = 1 \end{array}$$

## Some important sets (Numbers Sets)

### 1- Natural Numbers

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

An infinite set, closed under addition and multiplication.

### 2- Integer Numbers

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

An infinite set, closed under addition, multiplication and subtraction.

### 3- Rational Numbers

$$\mathbb{Q} = \{x: x = \frac{a}{b}; a, b \in \mathbb{Z}; b \neq 0\}$$

Closed set under addition, subtraction, multiplication and division operations. A rational number can either a finite decimal number such as (0.243) or a recurring number such as  $(0.\bar{3})$ .

### 4- Irrational Numbers

An irrational number is a type of real number which cannot be represented as a simple fraction

**Example:**  $\sqrt{2}$ ,  $\sqrt{3}$ ,  $\sqrt[3]{5}$ ,  $\pi = \frac{22}{7}$  are all irrational.

### 5- Real Numbers

$$\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^*$$

### 6- Complex Numbers

$$\mathbb{C} = x + iy; \quad i = \sqrt{-1}$$

where  $x$  is real part, and  $y$  is imaginary part.

✱ **In general**, we find that

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

- ❖  $\mathbb{Z}^+, \mathbb{Q}^+$  and  $\mathbb{R}^+$  are the sets of **positive integers, rationals** and **reals**, respectively. For example,  $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$
- ❖  $\mathbb{Z}^-, \mathbb{Q}^-$  and  $\mathbb{R}^-$  are the sets of **negative integers, rationals** and **reals**, respectively. For example,  $\mathbb{Z}^- = \{-1, -2, -3, \dots\}$
- ❖  $\mathbb{Z}^*, \mathbb{Q}^*$  and  $\mathbb{R}^*$  are the sets of **non-zero integers, rationals** and **reals**, respectively. For example,  $\mathbb{Z}^* = \{\pm 1, \pm 2, \pm 3, \dots\}$

## Operations on Sets

**(1) Union:** The set of elements that belong to either of two sets

$$A \cup B = \{x: x \in A \text{ Or } x \in B\}$$

**Example:**

If  $A = \{1,2,3,5\}$  and  $B = \{2,4,6\}$ , then  $A \cup B = \{1,2,3,4,5,6\}$

### Properties of Union

|                                             |                                                      |
|---------------------------------------------|------------------------------------------------------|
| (1) $A \cup A = A$                          | (2) $A \cup \phi = A$                                |
| (3) $A \cup U = U$                          | (4) $A \cup B = B \cup A$                            |
| (5) $A \cup (B \cup C) = (A \cup B) \cup C$ | (6) $A \subseteq (A \cup B), B \subseteq (A \cup B)$ |

**(2) Intersection:** The common elements of two sets

$$A \cap B = \{x: x \in A \text{ and } x \in B\}$$

**Example:**

If  $A = \{1,2,3,5\}$  and  $B = \{2,4,6\}$ , then  $A \cap B = \{2\}$

### Properties of Intersection

|                                             |                                                      |
|---------------------------------------------|------------------------------------------------------|
| (1) $A \cap A = A$                          | (2) $A \cap \phi = \phi$                             |
| (3) $A \cap U = A$                          | (4) $A \cap B = B \cap A$                            |
| (5) $A \cap (B \cap C) = (A \cap B) \cap C$ | (6) $(A \cap B) \subseteq A, (A \cap B) \subseteq B$ |

**Note:** Given sets  $A$  and  $B$ , they are said to be **disjoint** if and only if  $A \cap B = \phi$

**(3) Difference:** The set of elements that belong to a set but not to another

$$A - B = \{x: x \in A \text{ and } x \notin B\}$$

**Example:**

If  $A = \{1,5,6,12,20\}$  and  $B = \{3,6,12,18,20\}$ , then  $A - B = \{1,5\}$

### Properties of Difference

|                         |                                                  |
|-------------------------|--------------------------------------------------|
| (1) $A - A = \phi$      | (2) $A - \phi = A$                               |
| (3) $A - U = \phi$      | (4) $A - B \neq B - A$                           |
| (5) $A - B = A \cap B'$ | (6) $A - B = \phi \leftrightarrow A \subseteq B$ |

**(4) Symmetric Difference:** Given two sets, their symmetric difference is the set of elements that belong to either one or the other set but not both, denoted by  $A \Delta B$  or  $A \ominus B$  or  $A \oplus B$ .

$$A \oplus B = (A - B) \cup (B - A)$$

$$A \oplus B = (A \cup B) - (A \cap B)$$

**Example:**

If  $A = \{2,3,4,5\}$  and  $B = \{2,4,a,b\}$ , then  $A \oplus B = \{3,5,a,b\}$

### Properties of Symmetric Difference

|                                                     |                                               |
|-----------------------------------------------------|-----------------------------------------------|
| (1) $A \oplus A = \phi$                             | (2) $A \oplus \phi = A$                       |
| (3) $A \oplus U = \phi$                             | (4) $A \oplus B = B \oplus A$                 |
| (5) $A \oplus (B \oplus C) = (A \oplus B) \oplus C$ | (6) $A \oplus B = \phi \leftrightarrow A = B$ |

**(5) Complement Set:** The set of elements (in the universal set) that do not belong to a given set, denoted by  $\bar{A}$  or  $A^c$  or  $A'$

$$\bar{A} = \{x: x \in U \text{ and } x \notin A\}$$

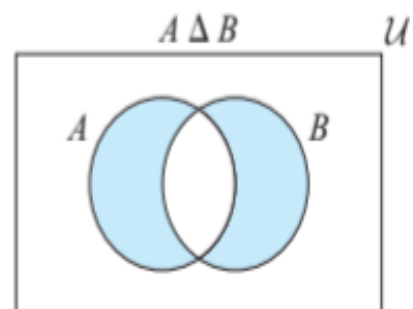
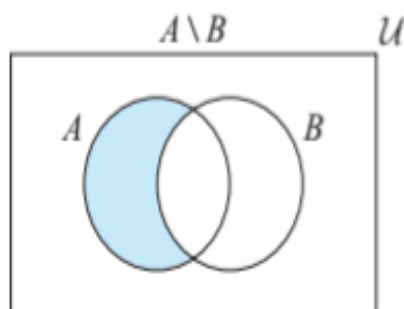
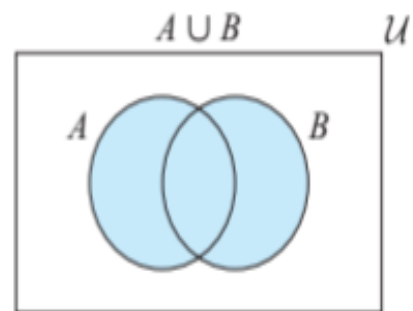
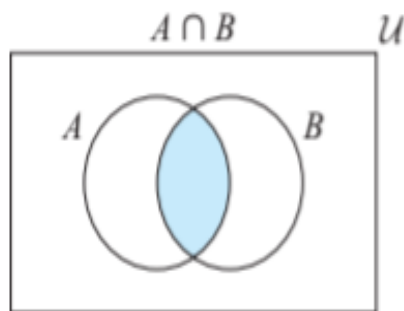
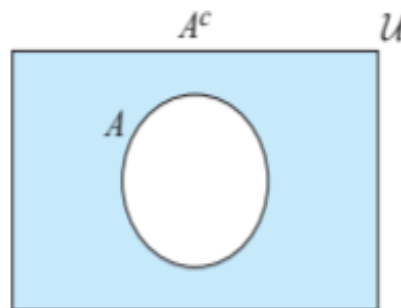
**Example:**

If  $A = \{1,2,3\}$  and  $U = \{1,2,3,4, \dots\}$ , then  $\bar{A} = \{4,5,6, \dots\}$

### Properties of Complement Set

|                                |                                                           |
|--------------------------------|-----------------------------------------------------------|
| (1) $\bar{\bar{A}} \cup A = U$ | (2) $\bar{A} \cap A = \phi$                               |
| (3) $\bar{\phi} = U$           | (4) $\bar{U} = \phi$                                      |
| (5) $\bar{\bar{A}} = A$        | (6) $A \subseteq B \rightarrow \bar{A} \supseteq \bar{B}$ |

### Venn Diagrams for Set Operations



## Laws of Algebra of Sets

### 1- Idempotent Laws

$$(i) \quad A \cup A = A$$

$$(ii) \quad A \cap A = A$$

### 2- Commutative Laws

$$(i) \quad A \cup B = B \cup A$$

$$(ii) \quad A \cap B = B \cap A$$

### 3- Associative Laws

$$(i) \quad (A \cup B) \cup C = A \cup (B \cup C)$$

$$(ii) \quad (A \cap B) \cap C = A \cap (B \cap C)$$

### 4- Distributive Laws

$$(i) \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(ii) \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

### 5- DeMorgan's laws

$$(i) \quad (A \cup B)' = A' \cap B'$$

$$(ii) \quad (A \cap B)' = A' \cup B'$$

### 6- Complement Laws

$$(i) \quad A \cup A' = U$$

$$(ii) \quad A \cap A' = \emptyset$$

$$(iii) \quad U' = \emptyset$$

$$(iv) \quad \emptyset' = U$$

### 7- Identity Laws

$$(i) \quad A \cup U = U \quad A \cap U = A$$

$$(ii) \quad A \cup \emptyset = A \quad A \cap \emptyset = \emptyset$$

### 8- Absorption Laws

$$(i) \quad A \cup (A \cap B) = A$$

$$(ii) \quad A \cap (A \cup B) = A$$

### 9- Set difference law

$$A - B = A \cap B'$$

### More laws of algebra of sets:

- For any **two finite sets** A and B;
  - (i)  $A - B = A \Leftrightarrow A \cap B = \emptyset$
  - (ii)  $(A - B) \cup B = A \cup B$
  - (iii)  $(A - B) \cap B = \emptyset$
  - (iv)  $A \subseteq B \Leftrightarrow \bar{B} \subseteq \bar{A}$
- For any **three finite sets** A, B and C;
  - (i)  $A \cap (B - C) = (A \cap B) - (A \cap C)$
  - (ii)  $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$

## Methods of Proof for Sets

### 1- Proof using Set-membership Tables

Let  $A$  be a subset of a universal set  $U$  and let  $u \in U$ . To use this method, we note that exactly one of the following is true:  $u \in A$  or  $u \notin A$ . Denote the situation where  $u \in A$  by **1 (membership)** and that where  $u \notin A$  by **0 (non-membership)**.

**Example:** Use **membership tables** to prove that  $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$

**Proof**

| A | B | $\bar{A}$ | $\bar{B}$ | $A \cup B$ | $\overline{(A \cup B)}$ | $\bar{A} \cap \bar{B}$ |
|---|---|-----------|-----------|------------|-------------------------|------------------------|
| 1 | 1 | 0         | 0         | 1          | 0                       | 0                      |
| 1 | 0 | 0         | 1         | 1          | 0                       | 0                      |
| 0 | 1 | 1         | 0         | 1          | 0                       | 0                      |
| 0 | 0 | 1         | 1         | 0          | 1                       | 1                      |

Since the **two columns corresponding** to  $\overline{(A \cup B)}$  and  $\bar{A} \cap \bar{B}$  are the identical, then the two sets are equal.

**Example:** Use **membership tables** to prove that  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

**Proof**

| A | B | C | $A \cup B$ | $A \cup C$ | $B \cap C$ | $A \cup (B \cap C)$ | $(A \cup B) \cap (A \cup C)$ |
|---|---|---|------------|------------|------------|---------------------|------------------------------|
| 1 | 1 | 1 | 1          | 1          | 1          | 1                   | 1                            |
| 1 | 1 | 0 | 1          | 1          | 0          | 1                   | 1                            |
| 1 | 0 | 1 | 1          | 1          | 0          | 1                   | 1                            |
| 1 | 0 | 0 | 1          | 1          | 0          | 1                   | 1                            |
| 0 | 1 | 1 | 1          | 1          | 1          | 1                   | 1                            |
| 0 | 1 | 0 | 1          | 0          | 0          | 0                   | 0                            |
| 0 | 0 | 1 | 0          | 1          | 0          | 0                   | 0                            |
| 0 | 0 | 0 | 0          | 0          | 0          | 0                   | 0                            |

Since the **two columns corresponding** to  $A \cup (B \cap C)$  and  $(A \cup B) \cap (A \cup C)$  are the identical, then the two sets are equal.

### 2- Proof using the Laws of Algebra of Sets

**Example:** Use the laws to algebra of set to prove that  $A \cup (\bar{A} \cap B) = A \cup B$

**Proof**

LHS

$$\begin{aligned} A \cup (\bar{A} \cap B) &= (A \cup \bar{A}) \cap (A \cup B) \\ &= U \cap (A \cup B) = A \cup B \end{aligned} \quad \#$$

**Example:** Use the laws to algebra of set to prove that  $A - (A - B) = A \cap B$

**Proof**

LHS

$$\begin{aligned} A - (A - B) &= A - (A \cap \bar{B}) \\ &= A \cap \overline{(A \cap \bar{B})} = A \cap (\bar{A} \cup B) \\ &= (A \cap \bar{A}) \cup (A \cap B) = \emptyset \cup (A \cap B) \\ &= A \cap B \end{aligned} \quad \#$$

## Inequalities

Inequalities are **mathematical expressions** involving the symbols  $>$ ,  $<$ ,  $\geq$  and  $\leq$ . It is used most often to **compare two numbers** on the number line by their size.

There are several different notations used to represent different kinds of inequalities:

- The notation  $a < b$  means that **a is less than b**
- The notation  $a > b$  means that **a is greater than b**
- The notation  $a \leq b$  means that **a is less than or equal to b**
- The notation  $a \geq b$  means that **a is greater than or equal to b**
- The notation  $a \ll b$  means that **a is much less than b**
- The notation  $a \gg b$  means that **a is much greater than b**.

## Properties on the number line

If  $a$ ,  $b$  and  $c$  are real numeres, then

- (1)  $a < b \rightarrow a \pm c < b \pm c$
- (2)  $a < b$  and  $c > 0 \rightarrow ac < bc$
- (3)  $a < b$  and  $c < 0 \rightarrow bc < ac$
- (4)  $a < b \rightarrow -b < -a$
- (5)  $a < b \rightarrow \frac{1}{b} < \frac{1}{a}$
- (6)  $a > 0 \rightarrow \frac{1}{a} > 0$

All of the previous properties are true if we put  $\leq$  instead of  $<$

## Intervals

An **interval in math** is measured in terms of numbers. An interval includes all the numbers that come between two particular numbers.

## Types of Intervals

| Interval Notation | Set Notation                 | Graph                                                                                | Type                           |
|-------------------|------------------------------|--------------------------------------------------------------------------------------|--------------------------------|
| $[a, b]$          | $\{x \mid a \leq x \leq b\}$ |  | Closed                         |
| $(a, b)$          | $\{x \mid a < x < b\}$       |  | Open                           |
| $[a, b)$          | $\{x \mid a \leq x < b\}$    |  | Half-open<br>or<br>half-closed |
| $(a, b]$          | $\{x \mid a < x \leq b\}$    |  |                                |



**Example:**

Solve the following inequalities and write the solution set using intervals on a number line

(1)  $2x - 1 < x + 3$

**Solution**

$$2x - x < 3 + 1 ;$$

$$x < 4 ;$$

$$\therefore x \in (-\infty, 4)$$

(2)  $\frac{-x}{3} < 2x + 1$

**Solution**

$$-x < 6x + 3 ;$$

$$-3 < 7x ;$$

$$x > \frac{-3}{7} ;$$

$$\therefore x \in \left(\frac{-3}{7}, \infty\right)$$

## Absolute Value

For any real number  $x$ , the **absolute value** or **modulus** of  $x$  is denoted by  $|x|$ , and is defined as

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

The absolute value of  $x$  is thus always either **a positive number or zero**, but never negative.

$$|3| = 3 ; \quad |0| = 0 ; \quad |-5| = -(-5) = 5 ; \quad |-|3|| = |3| = 3$$

### Properties of the absolute value

if  $x, y$  are real numbers, then

(1)  $|x| = 0$  iff  $x = 0$

(2)  $|xy| = |x||y|$

(3)  $\left|\frac{x}{y}\right| = \frac{|x|}{|y|} ; y \neq 0$

(4)  $|x + y| \leq |x| + |y|$

(5)  $|x|^2 = x^2$

### Definition

If  $a$  is a positive real number,  $x$  is any real number, then

$$|x| = a \quad \text{if and only if} \quad x = a \quad \text{or} \quad x = -a$$

$$|x| < a \quad \text{if and only if} \quad -a < x < a$$

$$|x| \leq a \quad \text{if and only if} \quad -a \leq x \leq a$$

$$|x| > a \quad \text{if and only if} \quad x > a \quad \text{or} \quad x < -a$$

$$|x| \geq a \quad \text{if and only if} \quad x \geq a \quad \text{or} \quad x \leq -a$$

**Example:** Find the solution set for the following inequalities

$$(1) |2x - 3| = 7$$

**Solution**

$$2x - 3 = 7; \quad 2x = 10 \quad ; \quad x = 5$$

**or**

$$2x - 3 = -7; \quad 2x = -4 \quad ; \quad x = -2$$

$$(2) |x - 5| \leq 3$$

**Solution**

$$-3 \leq x - 5 \leq 3; \quad 2 \leq x \leq 8; \quad \therefore x \in [2, 8]$$

$$(3) \frac{5x}{x-4} \geq 6$$

**Solution**

In this case we take **two cases** for **the denominator**  $x - 4$

✱ **The first case**  $x - 4 > 0$  or  $x > 4$

$$5x \geq 6(x - 4); \quad 5x \geq 6x - 24; \quad x \leq 24 \quad \therefore$$

$$4 < x \leq 24; \quad x \in (4, 24]$$

✱ **The second case**  $x - 4 < 0$  or  $x < 4$

$$5x \leq 6(x - 4); \quad 5x \leq 6x - 24; \quad x \geq 24$$

This is a contradiction  $x < 4$

$\therefore$  The solution set is  $\phi$

$\therefore$  The general solution set is the union of the two sets

$$(4, 24] \cup \phi = (4, 24]$$

$$(4) \left| \frac{2x-5}{x-6} \right| < 3$$

**Solution**

$$-3 < \frac{2x-5}{x-6} < 3$$

In this case we take **two cases** for **the denominator**  $x - 6$

✱ **The first case**  $x - 6 > 0$  or  $x > 6$

$$-3(x - 6) < 2x - 5 < 3(x - 6)$$

$$-3x + 18 < 2x - 5 < 3x - 18$$

$$\text{LHS} \quad -3x + 18 < 2x - 5; \quad 23 < 5x; \quad \therefore \frac{23}{5} < x$$

$$\text{RHS} \quad 2x - 5 < 3x - 18; \quad \therefore 13 < x$$

$\therefore$  The solution set is  $(13, \infty)$

✱ **The second case**  $x - 6 < 0$  or  $x < 6$

$$-3(x - 6) > 2x - 5 > 3(x - 6)$$

$$-3x + 18 > 2x - 5 > 3x - 18$$

$$\text{LHS} \quad -3x + 18 > 2x - 5; \quad 23 > 5x; \quad \therefore \frac{23}{5} > x$$

$$\text{RHS} \quad 2x - 5 > 3x - 18; \quad \therefore 13 > x$$

$\therefore$  The solution set is  $\left(-\infty, \frac{23}{5}\right)$

$\therefore$  The general solution set is all numbers except those in the period  $\left(\frac{23}{5}, 13\right)$

**Exercise**

- 1- Given that the universal set is  $U = \{1, 2, 3, \dots, 9\}$  and  $A = \{1, 2, 4, 7, 9\}$   
 $B = \{2, 3, 7, 8, 9\}$ ,  $C = \{3, 5, 8\}$ . Find the following operations

$$\begin{array}{lll} (1) \bar{A} & (2) A \cap B & (3) \overline{(A \cup B)} \\ (4) A - (B - C) & (5) A \oplus B & (6) \mathcal{P}(\bar{B}) \end{array}$$

- 2- Use the properties of Sets to prove the following operations

$$\begin{array}{l} (1) (A - B) \cap (B - A) = \emptyset \\ (2) (A - B) \cap B = \emptyset \\ (3) (A - B) \cap (A \cap B) = \emptyset \\ (4) \bar{A} \cap (A \cup \bar{B}) = \bar{A} \cap \bar{B} \\ (5) (A \cup B) \cap (A \cup \bar{B}) = A \end{array}$$

- 3- Prove the following operations

$$\begin{array}{l} (1) A = B \rightarrow A - B = \emptyset \\ (2) A = \bar{B} \rightarrow A \cap B = \emptyset \\ (3) B \subset A \rightarrow A \cup (A \cap B) = A \\ (4) B \subset A \rightarrow A \cap (A \cup B) = A \\ (5) A - \bar{B} = B - \bar{A} \\ (6) \overline{(\bar{A} \cap \bar{B})} = A \cup B \end{array}$$

- 4- Solve the following inequalities and write the solution set using intervals notation.

$$\begin{array}{lll} (1) \frac{6-x}{4} < \frac{3x-4}{2} & (2) \frac{4}{5}(x-2) < \frac{1}{3}(x-6) & (3) \frac{2x-5}{4} \leq 3 \\ (4) 3(2-x) > 2(3+x) & (5) x+3(2-x) \geq 4-x & (6) 8-3x \geq 5 \\ (7) \left| \frac{3x}{5} - 1 \right| > \frac{2}{5} & (8) |2-3x| > 5 & (9) \left| \frac{x}{2} - 1 \right| = 1 \\ (10) \left| 3 - \frac{1}{x} \right| < \frac{1}{2} & (11) \left| \frac{2}{x} - 4 \right| < 3 & (12) |8-3x| = \frac{9}{2} \\ (13) |2x-3| \geq 1 & (14) \left| \frac{x+1}{2} \right| \geq 1 & (15) \left| \frac{3}{2}x - 1 \right| \leq 2 \\ (16) \left| \frac{x}{5} - 1 \right| \leq 1 \end{array}$$